

BINARY NUMBERS – SHORT REVISION NOTES

Understand Level (Explain Ideas & Concepts)

1. Understanding the Binary Number System

Concept	Explanation	Example
Definition	A number system that uses only two digits (0 and 1)	1011_2
Base	The number of digits used = 2	Base 2
Why Binary?	Computers use two states: ON (1) and OFF (0)	1 = current ON, 0 = current OFF
Place Values	Powers of 2: 1, 2, 4, 8, 16, 32...	$2^0, 2^1, 2^2, 2^3...$

Understanding the Concept:

Decimal System (Base 10):

Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 (10 digits)

Place Values: 1, 10, 100, 1000... (Powers of 10)

Binary System (Base 2):

Digits: 0, 1 (2 digits)

Place Values: 1, 2, 4, 8, 16, 32... (Powers of 2)

Visual Understanding:

Decimal: 3 7 2 5
Places: 10^3 10^2 10^1 10^0
Value: $3000 + 700 + 20 + 5 = 3725$

Binary: 1 1 0 1
Places: 2^3 2^2 2^1 2^0
Value: 8 + 4 + 0 + 1 = 13

Key Understanding: Binary is just like decimal, but with only two digits and place values that are powers of 2 instead of powers of 10.

2. Understanding Place Values in Binary

Position	Place Value	Meaning	Example: 1011_2
1st (rightmost)	$2^0 = 1$	Counts ones	$1 \times 1 = 1$
2nd	$2^1 = 2$	Counts twos	$1 \times 2 = 2$
3rd	$2^2 = 4$	Counts fours	$0 \times 4 = 0$
4th	$2^3 = 8$	Counts eights	$1 \times 8 = 8$
5th	$2^4 = 16$	Counts sixteens	-
6th	$2^5 = 32$	Counts thirty-twos	-

Understanding with Abacus:

Abacus for 1101_2 :

8 4 2 1

■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■

■ ● ■ ● ■ ■ ■ ● ■ (1×8 + 1×4 + 0×2 + 1×1)

■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■

1 1 0 1

Abacus for 1101_2 :

8 4 2 1

■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■

■ ● ■ ● ■ ■ ■ ● ■ (1×8 + 1×4 + 0×2 + 1×1)

■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■

1 1 0 1

Abacus for 1101_2 :

8	4	2	1
■	■	■	■
■●	■●	■	■●
$(1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1)$			
■	■	■	■
1	1	0	1

Abacus for 1101_2 :

8	4	2	1
■	■	■	■
■●	■●	■	■●
$(1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1)$			
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Abacus for 1101_2 :

8	4	2	1
■	■	■	■
■●	■●	■	■●
$(1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1)$			
■	■	■	■
1	1	0	1

Key Understanding: Each position in binary represents a power of 2, just like each position in decimal represents a power of 10.

3. Understanding Why We Use Binary

Concept	Explanation	Example
Computer Systems	Computers use two states: ON/OFF	1 = ON, 0 = OFF
Data Storage	All data stored as 0s and 1s	Text, images, videos = binary
Lighting Systems	Current ON (1) or OFF (0)	Circuit design
Digital Instruments	Everything digital uses binary	Calculators, phones, computers

Understanding the Connection:

```
Real World → Digital World
ON   → 1
OFF  → 0
Yes  → 1
No   → 0
True → 1
False → 0
```

```
Real World → Digital World
ON   → 1
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Key Understanding: Binary is the language of computers because it's easy to represent two states (ON/OFF, True/False, 0/1).

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4. Understanding Conversion: Decimal to Binary

Method: Repeated Division by 2

Step	Why?	Example: $22_{10} \rightarrow 10110_2$
1	Divide by 2 to find the 1st digit (remainder)	$22 \div 2 = 11 \text{ R } 0$
2	Continue dividing the quotient	$11 \div 2 = 5 \text{ R } 1$
3	Continue until quotient is 0	$5 \div 2 = 2 \text{ R } 1$
4	Continue	$2 \div 2 = 1 \text{ R } 0$
5	Continue	$1 \div 2 = 0 \text{ R } 1$
6	Read remainders from BOTTOM to TOP	10110_2

Understanding the Process:

$$\begin{aligned} 22 &= 16 + 4 + 2 \\ &= 2^4 + 2^2 + 2^1 \\ &= 1 \times 16 + 0 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1 \\ &= 10110_2 \end{aligned}$$

We got the same answer! ✓

Key Understanding: The remainders from division by 2 give the binary digits from right to left.

5. Understanding Conversion: Binary to Decimal

Method: Multiply by Powers of 2

Step	Why?	Example: $1101_2 \rightarrow 13_{10}$
1	Identify each digit's position	1, 1, 0, 1
2	Assign powers of 2 (from right)	$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
3	Calculate each term	$1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1$
4	Add all terms	$8 + 4 + 0 + 1 = 13$

Visual Understanding:

$$\begin{aligned} 1101_2 &= 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 \\ &= 8 + 4 + 0 + 1 \\ &= 13 \end{aligned}$$

This is like reading a decimal number:

$$3725_{10} = 3 \times 1000 + 7 \times 100 + 2 \times 10 + 5 \times 1$$

Key Understanding: Each binary digit is multiplied by its place value (power of 2), just like decimal digits are multiplied by powers of 10.

6. Understanding Binary Addition

Rules:

Rule	Why?	Example
$0 + 0 = 0$	No ones to add	$0_2 + 0_2 = 0_2$
$0 + 1 = 1$	One one	$0_2 + 1_2 = 1_2$
$1 + 0 = 1$	One one	$1_2 + 0_2 = 1_2$
$1 + 1 = 10_2$	Two ones = one two (carry 1)	$1_2 + 1_2 = 10_2$

Understanding Carrying:

$$\begin{array}{r} 101_2 \\ + 10_2 \\ \hline 111_2 \end{array}$$

Why?

Right column: $1 + 0 = 1$

Middle column: $0 + 1 = 1$

Left column: $1 + 0 = 1$

$$\begin{array}{r} 101_2 \\ + 1_2 \\ \hline 110_2 \end{array}$$

Why?

Right column: $1 + 1 = 0$ (carry 1 to next column)

Middle column: $0 + 0 + \text{carry } 1 = 1$

Left column: $1 + 0 = 1$

Visual with Abacus:

$$101_2 + 10_2 = 111_2$$

4	2	1	4	2	1	4	2	1
■ ■ ■ ■ ■ ■ ■ ■			■ ■ ■ ■ ■ ■ ■ ■			■ ■ ■ ■ ■ ■ ■ ■		
●	■	■	●	■	■	+	■	■
=			■	●	■	■	■	■
■ ■ ■ ■ ■ ■ ■ ■			■ ■ ■ ■ ■ ■ ■ ■			■ ■ ■ ■ ■ ■ ■ ■		
1	0	1	0	1	0	1	1	1

Key Understanding: Binary addition follows the same principles as decimal addition — carry when the sum exceeds the base (2 instead of 10).

7. Understanding Binary Subtraction

Rules:

Rule	Why?	Example
$0 - 0 = 0$	Nothing to subtract	$0_2 - 0_2 = 0_2$
$1 - 0 = 1$	One minus nothing	$1_2 - 0_2 = 1_2$
$1 - 1 = 0$	One minus one	$1_2 - 1_2 = 0_2$
$0 - 1 = \text{Borrow from left}$	Need to borrow 2 from next column	$10_2 - 1_2 = 1_2$

Understanding Borrowing:

```
  1102
-   12
■■■■■
  1012
```

Why?

Right column: $0 - 1 \rightarrow$ Borrow 1 from middle column

Middle column becomes 0, right column becomes 2

$2 - 1 = 1$

Middle column: $0 - 0 = 0$

Left column: $1 - 0 = 1$

```
  1002
-   12
■■■■■
   112
```

Why?

Right column: $0 - 1 \rightarrow$ Borrow from middle (0) $\rightarrow$ borrow from left

Left column becomes 0, middle becomes 2

Borrow from middle $\rightarrow$ middle becomes 1, right becomes 2

$2 - 1 = 1$

Middle column: $1 - 0 = 1$

Left column: $0 - 0 = 0$

Key Understanding: Borrowing in binary is similar to decimal — you borrow 1 from the next column, which equals 2 in binary.

8. Understanding the Relationship: Powers of 2

Concept	Explanation	Example
2^0	The first place value	1
2^1	The second place value	2
2^2	The third place value	4
2^3	The fourth place value	8
2^4	The fifth place value	16
2^5	The sixth place value	32

Understanding the Pattern:

$$\begin{aligned}2^0 &= 1 & (1) \\2^1 &= 2 & (1 \times 2) \\2^2 &= 4 & (2 \times 2) \\2^3 &= 8 & (4 \times 2) \\2^4 &= 16 & (8 \times 2) \\2^5 &= 32 & (16 \times 2) \\2^6 &= 64 & (32 \times 2) \\2^7 &= 128 & (64 \times 2) \\2^8 &= 256 & (128 \times 2)\end{aligned}$$

Pattern: Each power is double the previous one!

Key Understanding: Powers of 2 double each time, just like powers of 10 multiply by 10 each time.

9. Understanding the Relationship: Binary and Decimal

Binary	Decimal	Understanding
1_2	1_{10}	$1 \times 1 = 1$
10_2	2_{10}	$1 \times 2 + 0 \times 1 = 2$
11_2	3_{10}	$1 \times 2 + 1 \times 1 = 3$
100_2	4_{10}	$1 \times 4 + 0 \times 2 + 0 \times 1 = 4$
101_2	5_{10}	$1 \times 4 + 0 \times 2 + 1 \times 1 = 5$
110_2	6_{10}	$1 \times 4 + 1 \times 2 + 0 \times 1 = 6$
111_2	7_{10}	$1 \times 4 + 1 \times 2 + 1 \times 1 = 7$
1000_2	8_{10}	$1 \times 8 + 0 \times 4 + 0 \times 2 + 0 \times 1 = 8$

Key Understanding: Binary numbers follow the same pattern as decimal numbers, but with base 2 instead of base 10.

10. Understanding Applications of Binary

Application	Explanation	Example
Computers	Store data as 0s and 1s	Text = binary code
Calculators	Process numbers in binary	All calculations done in binary
Lighting Circuits	ON/OFF states	1 = ON, 0 = OFF
Digital Devices	All digital systems use binary	Phones, tablets, TVs

Understanding:

In a computer:
1 = Current ON (electricity flowing)
0 = Current OFF (no electricity)

Text in binary:

A = 01000001_2

B = 01000010_2

C = 01000011_2

Key Understanding: Binary is used because it's simple — only two states to represent, making electronics easy to design.

11. Common Misunderstandings to Avoid

Misunderstanding	Why It's Wrong	Correct Understanding
"Binary has the digit 2"	Binary only has 0 and 1	$10_2 = 2_{10}$
" $10_2 = 10_{10}$ "	$10_2 = 2_{10}$, not 10_{10}	$10_2 = 2$, $10_{10} = 1010_2$
" $1+1=0$ "	$1+1=10_2$ (carry 1)	$1_2+1_2=10_2$
"Binary is harder than decimal"	It's just different	Both systems follow same rules with different bases
"Computers only use 0s and 1s"	All data is binary	The computer converts binary to everything we see

12. Check Your Understanding

No.	Question	Explain
1	What is the base of binary?	2 (uses two digits: 0 and 1)
2	What are the place values in binary?	Powers of 2: 1, 2, 4, 8, 16...
3	Why do computers use binary?	Easy to represent ON/OFF states
4	How do you convert decimal to binary?	Repeated division by 2
5	How do you convert binary to decimal?	Multiply by powers of 2
6	What is $1_2 + 1_2$?	10_2 (carry 1 to next column)
7	What is $0_2 - 1_2$?	Need to borrow from the next column
8	Why is $2^0 = 1$?	Any number to power 0 equals 1
9	What is the 5th place value in binary?	16 (2^4)
10	Why is binary important?	Used in all digital devices

13. Key Understandings to Remember

- **BINARY = BASE 2** (uses digits 0 and 1)
- **PLACE VALUES = POWERS OF 2**
 $2^0=1, 2^1=2, 2^2=4, 2^3=8, 2^4=16\dots$
- **DECIMAL → BINARY: DIVIDE BY 2 REPEATEDLY**
Read remainders from bottom to top
- **BINARY → DECIMAL: MULTIPLY BY POWERS OF 2**
Add all results together
- **BINARY ADDITION:**
 $1_2 + 0_2 = 1_2$
 $1_2 + 1_2 = 10_2$ (carry 1)
- **BINARY SUBTRACTION:**
If $0 - 1$, borrow from next column
Borrowed 1 = 2 in the current column
- **COMPUTERS USE BINARY BECAUSE:**
Only two states: ON (1) and OFF (0)
Easy to implement in electronics
- **0 IS NEVER A SUBSCRIPT** (2_0 is wrong!)
Always write: 2 (as in 101_2)

Revision Tip: Visualize binary numbers on an abacus. Understanding place values is the key to understanding binary! Remember: binary works exactly like decimal — just with base 2 instead of base 10.